

Digital Puppetry: Utilizing Extended Reality Technologies for Animations

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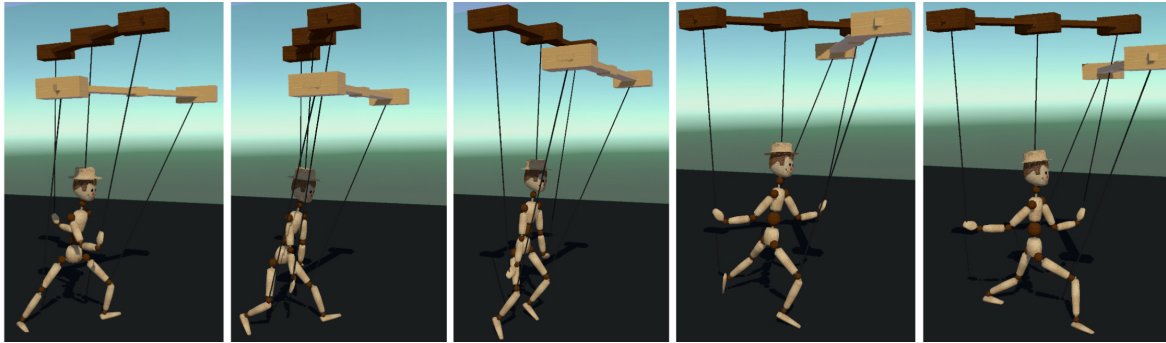


Figure 1: Image sequence of an animated walk cycle.

ABSTRACT

In this work, we present an interaction design for a digital puppet in a virtual reality application. The prototype is inspired by the concept of analog puppetry and implemented in *Unity*¹ for the Quest and Quest 2. All limbs of the puppet can be animated from different perspectives with the controllers. Our approach may benefit game designers and researchers who seek to use puppetry as a game element or want to explore possibilities of puppetry in virtual reality for authoring tools.

Keywords: Puppetry, virtual reality, interaction design.

Index Terms: Human-centered computing—Interaction design

1 INTRODUCTION

Character animation is a specialized area of the animation process, which is widely used today in the fields of film, game, advertising, education, and the metaverse. The classical approach of animating is done with the help of 3D authoring tools that include keyframe editors. However, creating complex and simultaneous movements for a character can be complicated. Especially, for beginners, it is difficult to produce compelling animations that include all parts of the body, and even for professionals, it is challenging to convey an idea for animation in three-dimensional space without having to sketch a 2D animation on a storyboard.

With the increase of technological advancement and commercial popularity of Virtual Reality (VR) head-mounted displays, additional opportunities for creating animations in an immersive three-dimensional environment arise. Still, typical interfaces for keyframe editors cannot be fully transferred from a two-dimensional screen to a three-dimensional virtual environment. The key for believable animations is to fully understand how to tell stories by manipulating

all parts of the body. Even though there exist different types of motion capture studios nowadays, they are not applicable for every character animation and they are still expensive for small studios. This work focuses on bringing the long tradition of the marionette, which is also called string puppet in France and manipulated from above by strings or threads attached to a control, into virtual reality to investigate possible interaction design choices for digital puppets that could help build animation authoring tools for beginners in the future.

2 RELATED WORK

Several recent studies have informed progress on animating puppets in VR for games or animation authoring tools. Nitsche and McBride [3] established a puppet-based perspective to character controls for VR and highlighted the emerging design space for games. However, there were numerous challenges that a standard VR interface cannot fully address. One of them regards the lack of haptic feedback. Nevertheless, design approaches in the project "Archiving Performative Objects" manage to open up and exemplify new design spaces for VR game interfaces. In a following paper, Nitsche and McBride [4] aimed at applying and conserving puppetry as a creative practice in VR in the project "Archiving Performative Objects". Their paper reports on the design and implementation of different puppet types in VR that are evaluated together with experts and non-experts of puppetry. Results demonstrate that the creative setup is successful, especially among experts. *AnimationVR* [6] is a written *Unity* (by Unity Technologies) plugin for the HTC Vive headset that enables users to animate characters with controllers in a puppeteering technique. Additionally, users can record the animation of different characters with an integrated timeline editor in VR. The result of the study with animation experts concludes that the workflow speed is increased with the drawback of lower animation precision.

Huang et. al. [1] developed a combat game in *Unreal Engine 4* (UE4, by Epic Games)² where it is possible to control the limbs of a two-dimensional humanoid puppet with hand gestures that are recognised by *Leap Motion*. The goal of the game is to hit the opponent and collect points. By introducing new hand gestures, the puppets

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¹<https://unity.com>

²<https://www.unrealengine.com>



Figure 2: Grab the wooden control bars.

can perform special tricks that deal more damage to the opponent. Husinsky and Bruckner [2] created a low-cost virtual studio setup where 2D characters can be animated with the *Leap Motion* as a tracking device. Physical people and live-animated characters can perform together in a virtual environment that is implemented in VVVV and the result can be watched both on a screen and in a head-mounted display. Velho and Lupiac [5] presented a framework for a gesture interaction system that is used to control 3D models of puppets with bare hands by using Leap Motion. Way et al. [7] presented a multi-user puppetry theater in VR where users could animate traditional chinese glove puppets with one hand by mapping each limb to individual fingers and detecting the movement with the *Leap Motion* sensor. The puppet show can be performed by two users and it is possible to record the complete performance. Our work is inspired by the concepts of Nitsche and McBride [4] since we also aim to bring analog marionette puppetry into the virtual environment. Our main contribution is to explore possibilities, challenges and limitations of digital marionettes in VR that could be beneficial for animation and games in future work.

3 INTERACTION DESIGN

We created a game environment that serves as a sandbox to get used to and experiment with the puppet's unique control mechanisms. The game setup enabled us to obtain a basic understanding of potentials of our ideas and to identify areas of improvement. In our approach, the user can navigate to the marionette in the scene by using the joystick of the left VR controller to interact with the puppet in play mode. A continuous movement was chosen as a type of locomotion since it would not be possible to create animations in which the whole puppet moves by teleporting from one position to another one.

The final interaction design of the puppet is heavily based on the way of how controlling marionettes in the real world work. In order to grab the wooden control bars of the marionette, the user can reach out for them and press the grip button of the controllers to hold them as shown in figure 2. With this design decision, the user can decide with which hand the controls for the legs or the hands should be

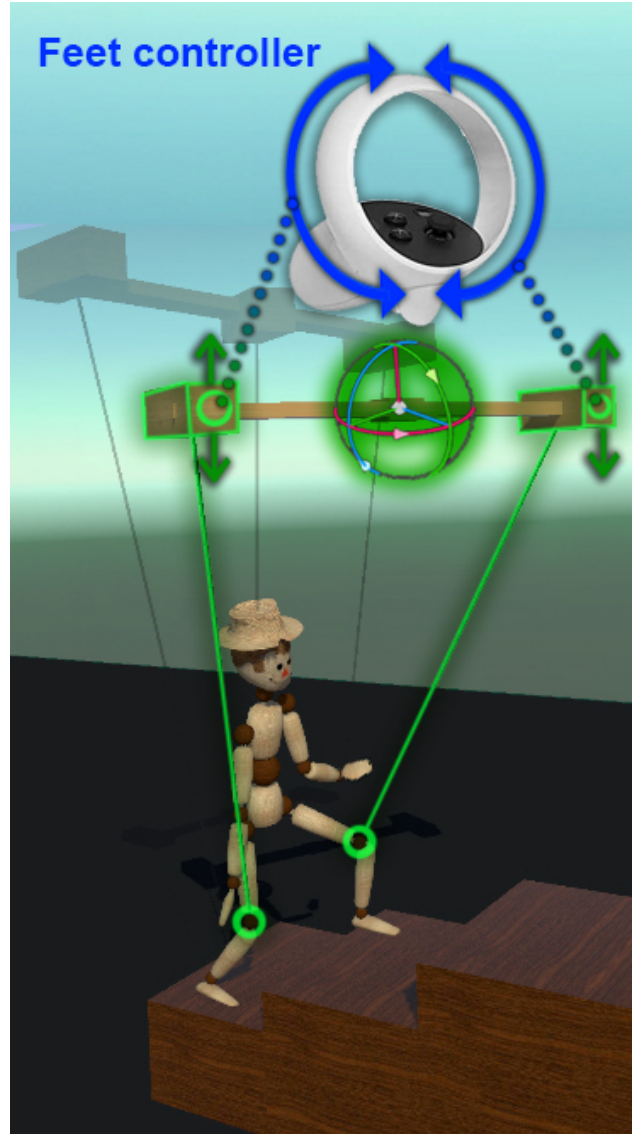


Figure 3: Feet controller.

grabbed.

One wooden control bar is attached to the arms and the head of the marionette and the other one is connected to the knees with strings. By lifting or lowering the bar for the arms, the general posture of the puppet can be influenced which makes it possible to animate the puppet in a standing, sitting, or kneeling position. By manipulating the height of the bar for the legs, both legs can be lifted or lowered together. While a horizontal rotation of the arm control bars leads to a turn of the whole upper body, a change of horizontal orientation of the leg controls leads to one leg being stretched forwardly depending on the direction of the rotation. Figure 3 shows how the rotation of the wooden control bars influences the position of the legs.

By lifting both controllers, a jump can be animated and with a combination of changing the height and the orientation of the leg controller, a simple walk cycle can be created as illustrated in figure 1. While gravity is applied to the body of the marionette, the wooden controls are not affected by any physics which makes it possible to easily release them and pick them up again from a different

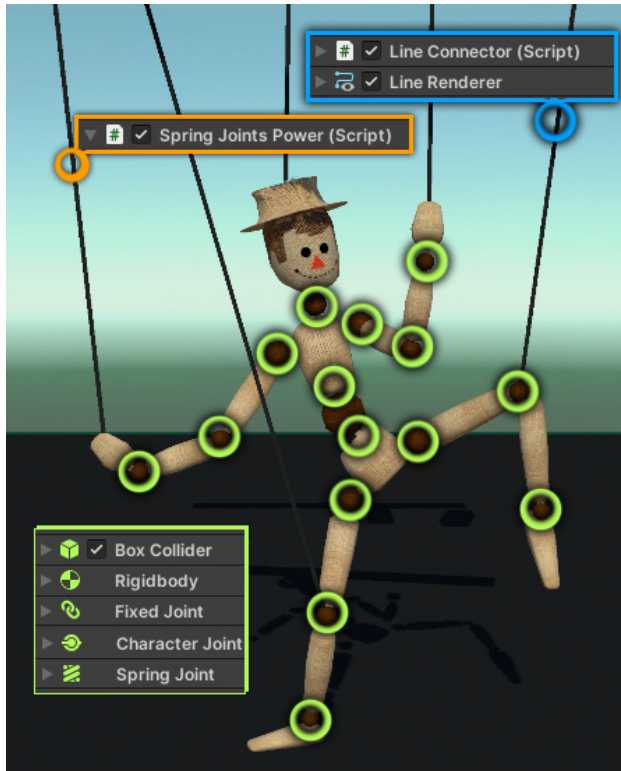


Figure 4: String puppet rig.

perspective. This decision aims to support the user since it is easier to look at the current posture from different angles and it prevents tangled strings that would happen with a released puppet in the real world.

4 TECHNICAL DESIGN

The project is implemented in the 2020.3.21f1 *Unity* version and built for the *Oculus Quest* and *Quest 2*³. To provide simple interactions like locomotion and grabbing the wooden control bars, the XR interaction toolkit is used. Since both headsets are standalone devices with two 6DoF controllers, no additional PC or console is required which is also the reason why we decided to choose both headsets as a target device. After correctly setting up the guardian system which is responsible for defining the boundaries of the play area for Oculus headsets, the prototype can be used at any possible location where it is at least possible to fully stretch both arms.

The virtual low-poly scarecrow puppet and the wooden controllers were modeled and textured in *Autodesk 3ds Max*⁴ 2022 and the final 3D model with the materials was imported in *Unity* as an FBX file. The character rig itself was built in *Unity* and each part of the puppet body was joined together with physics components like the Character Joint, Fixed Joint, and Hinge Joint. To achieve gravity, the rigid body component is also used. Following this, five Spring joints were attached to the marionette head, hands, and knees. These were connected to wooden controllers as well to rig the strings. Afterward, the Line connector script was written to render the strings in play mode by using the Line Renderer component. This is more optimized than including geometry for the strings as shown in figure 4. The "SeesawRotation" script was written to imitate the body

³<https://www.oculus.com>

⁴<https://www.autodesk.com>

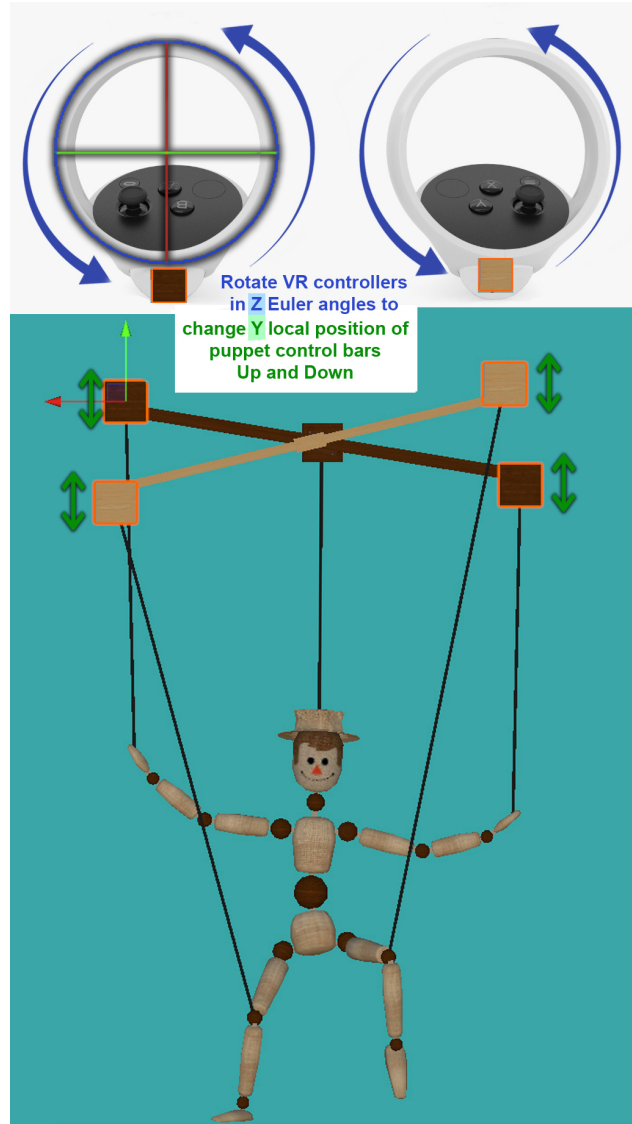


Figure 5: SeesawRotation script.

movement of a real-world marionette. The current position in the y-axis of the individual limbs is dependent on the degree of rotation of the control bars. By grabbing the wooden control bars and changing the rotation in the z-axis, individual arms and legs can be lifted or lowered as demonstrated in figure 5.

5 LESSONS LEARNED

Apart from internal testing with the team, we also performed informal tests with colleagues of the game design prototyping course of our university where this project originally started. These tests gave us a lot of useful feedback and helped us to improve the interaction design of the digital puppet.

In the early stages of designing, the puppet was rigged in *3Ds Max*. However, the rig was not reliant enough and the movement of the puppet did not feel natural with the bones system. Therefore, the whole rig was built with physic components in *Unity* itself which had a more satisfactory outcome. In other early approaches, the interaction for controlling the puppet was designed in a more abstract way by controlling the left arm and leg with the left controller

and the right arm and leg with the right controller. By mapping the activation of control to the primary and secondary buttons of the controller, the user should be able to control each limb individually and also decide when to release it again. We quickly realized that it is not intuitive to think about pressing four buttons individually and therefore, we focused on the implementation that is more related to analog marionette controls.

Another design idea was to let the head of the puppet follow the position and rotation of the headset. This worked very well but led to the problem that the user could not fully release the whole puppet and control it from a different perspective. For this reason, we decided to attach the head to the arm controller in our current implementation.

6 LIMITATIONS AND FUTURE WORK

We acknowledge that we have limitations in our work. Since our approach is still a work-in-progress, we recognize that a formal user study consisting of qualitative and quantitative methods would give important information on how users with different levels of knowledge in puppeteering or animating interact with the digital marionette. Future research will lead to two directions. In the first direction, a user study will be conducted that will compare different levels of bounciness for the strings of the marionette and the impact on the game experience. The main inspiration for this research direction comes from games like *QWOP*⁵, *Octodad*⁶ or *Goat Simulator*⁷ that include game physics that are difficult to predict and control, but still deliver a fun and challenging game experience. By comparing different levels of bounciness for the puppet's strings, we want to explore to what extent the game experience in VR games could benefit from these challenging physics. The second direction will focus on building a low budget animation-authoring system in VR for people who are still new in animating characters and cannot afford expensive software and hardware.

Another limitation of our current prototype becomes visible when the distance between the grabbed control bars becomes too high by fully stretching both arms. Since users can move their hands without limits and do not feel the stretching of strings like for a real-world marionette, extending these limits in our current implementation leads to undesired behavior. In order to solve this problem, we plan to restrict the possible movement of the controllers to the limit of the ragdoll. If the user wants to extend the wooden control bars further than the limit, the control bars will not follow the hands further and haptic feedback in terms of increasing vibration of the controllers will inform the user that the possible limit is reached.

7 CONCLUSION

In this paper, we presented a prototype that brings interaction concepts from an analog marionette in a virtual reality application. The digital marionette enables users to control and animate all limbs of the puppet from different perspectives with controllers. The result yields potential for a future low-budget animation authoring tool that especially supports beginners in the art of animation. The digital puppet could enable users to animate the puppet more intuitively than a classic authoring tool since the possibility of navigating around the character supports the spatial understanding and the controls of the puppet would allow users to get a better understanding of motion sequences of characters.

Furthermore, we would like to use the digital marionette in a future VR game and conduct a formal study to examine how the physics of the marionette's strings affect the game experience. This research direction is inspired by games that achieve to bring a fun and challenging game experience with unpredictable physics. By

comparing different levels of the bounciness of the puppet's strings in a VR game, we aim to explore the balance between a satisfying game experience and challenging physics. The lessons that were learned during this work may benefit game designers and researchers who also intend to bring analog puppetry into the immersive virtual environment.

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⁵<http://www.foddy.net/Athletics.html>

⁶<https://octodad.com/>

⁷<https://www.goat-simulator.com/>